Do Now · retrieval

Boolean logic, gates and truth tables

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the difference between a stack and a queue when removing items?

Q2

What must be true about an array before you use binary search?

Q3

For int[] numbers = {5, 9, 14, 18, 23}; , what value is stored at zero-based index 3?

Q4

A binary search looks for 23 in the sorted array [2, 7, 11, 16, 23, 31, 44] . Which zero-based indexes are checked, in order?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

AND — both inputs must be 1 — A gate takes an input (or inputs) and produces one output, Q . For AND, both inputs must be 1. — Example: 1 AND 0 = 0; 1 AND 1 = 1.

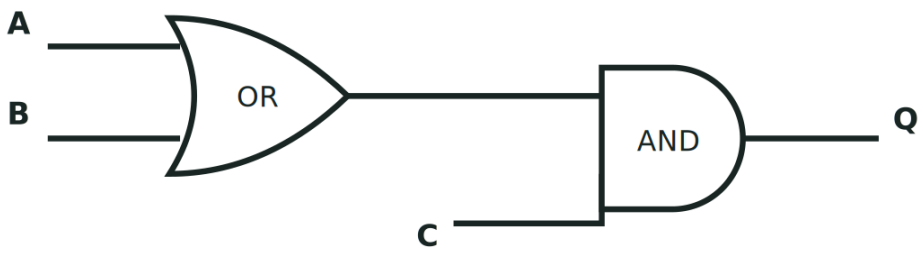
Core idea 2

NOT — reverse one input — NOT has just one input . The small circle on the output is the inversion symbol. — Example: NOT 1 = 0; NOT 0 = 1.

Core idea 3

XOR (⊻) — exactly one input is 1 — XOR is different from OR: when both inputs are 1, XOR gives 0. Notice the extra curved line at the input. — Symbol: Q = A ⊻ B. Compare: 1 ∨ 1 = 1, but 1 ⊻ 1 = 0.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Words first, then Boolean symbols — These symbols are shorthand for the rules you have just seen: — Say: A OR B .

Core idea 5

Check the circuit with a truth table — For three inputs there are 8 combinations . Work out the OR column first, then use that result with C. — CODE:

A B C | A∨B | Q
0 0 0 | 0 | 0
0 0 1 | 0 | 0
0 1 0 | 1 | 0
0 1 1 | 1 | 1
1 0 0 | 1 | 0
1 0 1 | 1 | 1
1 1 0 | 1 | 0
1 1 1 | 1 | 1

Vocabulary · use the correct technical terms

Boolean — A value with two states.

AND — True only if all required inputs are true.

OR — True if at least one input is true.

NOT — Reverses a Boolean value.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Secure door access

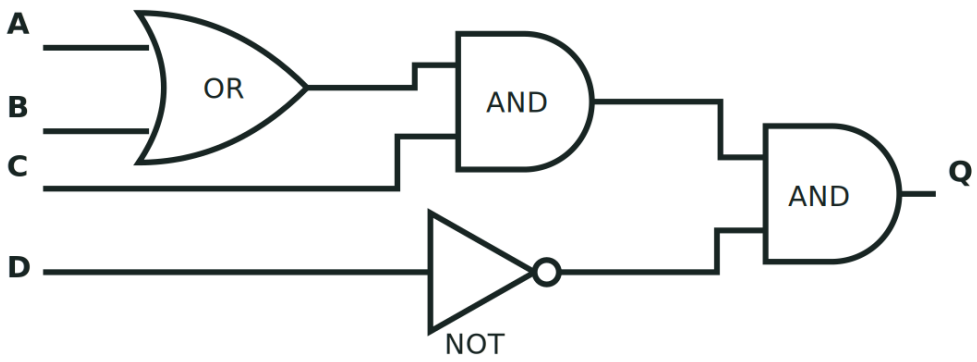
Follow the worked method

1. What switches ON? — CODE:
Q = door unlocks — Start with the result the question is asking us to control.

2. What must also be true? — CODE:
(A ∨ B) ∧ C — The building must be open , so join the options with AND.

3. Use intermediate columns — For Q = (A ∨ B) ∧ ¬C, three inputs give $2^3 = 8$ combinations. Write every input row, calculate A∨B, then ¬C, then combine those two columns with AND. — CODE:
A B C | A∨B | ¬C | Q
0 0 0 | 0 | 1 | 0
0 0 1 | 0 | 0 | 0
0 1 0 | 1 | 1 | 1
0 1 1 | 1 | 0 | 0
1 0 0 | 1 | 1 | 1
1 0 1 | 1 | 0 | 0
1 1 0 | 1 | 1 | 1
1 1 1 | 1 | 0 | 0 — Example row 010: A∨B = 1 and ¬C = 1; therefore Q = 1. Always complete the intermediate columns before Q.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Which gate?
Are there any alternative routes? What must be true? Write Q, draw the gate and predict the truth-table outputs.
CODE:
A B | Q
0 0 | ?
0 1 | ?
1 0 | ?
1 1 | ?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Burglar alarm circuit
A burglar alarm has four inputs. Each input is 1 when the condition is true and 0 otherwise:
A: door sensor detects movement
B: motion sensor detects movement
C: alarm is armed
D: test mode is enabled
The alarm sounds (Q = 1) only when at least one sensor detects movement , the alarm is armed , and test mode is off .
Your tasks: (1) Write the Boolean expression for Q. (2) Draw the logic-gate circuit using AND, OR and NOT gates. (3) Label A, B, C, D and Q.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Expression to gate plan

Convert Q = (A ⊻ B) ∧ ¬C into a gate-by-gate plan and complete its truth table.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A three-input Boolean problem has eight possible input combinations in its truth table.

Use key vocabulary from memory

1. Explain Boolean in your own words.

2. Explain AND in your own words.

3. Explain OR in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.